

Review and critique of the information systems development project failure literature: An argument for exploring information systems development project distress

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Abstract

High failure rates of information systems development (ISD) projects continue to trouble organizations and information systems practices. Such a state of affairs has been of great concern for the information systems discipline for decades, motivating information systems researchers to focus on and extensively study ISD project failure. However, the increasing complexity and uncertainty of ISD projects and contemporary system development processes are challenging ISD project failure scholarship. In this article, we ask the questions: What are the contributions and weaknesses of the extant ISD project failure/success literature? What are potential avenues to move the ISD literature forward? To answer these questions, first, we present a literature review that assesses research contributions within the major perspectives on ISD failure (i.e. rationalist, process and narrative). While the extant research within all perspectives make significant contributions to knowledge, we find that researchers remain preoccupied with ‘project failure’ as an end state of an ISD project. They pay little attention to problematic situations arising during ISD projects before they become failed projects. Based on the review and critique of the literature, we then argue that there is a significant benefit in extending research focus from ISD project failure to ‘ISD project distress’, which we define as a harmful project condition involving dynamic and fluid constellation of critical problems that are difficult to identify, understand and resolve. While ISD project distress is an increasingly perilous and consequential phenomenon, little is known about its nature and potential responses. Drawing from the sensemaking literature, we propose a multilevel theoretical framework for understanding the nature and sources of ISD project distress that provides a foundation for exploring early detection and timely response. We demonstrate the theoretical and practical relevance of the concept of ISD project distress and propose a corresponding research agenda.

Keywords

Information systems development (ISD), ISD project failure, ISD project distress, ISD project management, sensemaking, literature review, critique

Introduction

Information systems development (ISD) projects continue to fail at high rates (Doherty and King, 2001; Lee et al., 2014; Vaidya et al., 2013), often suggested to be 70% or higher (Doherty et al., 2012; Drummond, 2005; Hochstrasser and Griffiths, 1991; Keil and Mähring, 2010). Such high failure rates of ISD projects have motivated information

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systems (IS) researchers to focus on and keep exploring ISD project failure/success, contributing to the wealth of knowledge in the IS literature (e.g. Conboy, 2010; Doherty et al., 2012; Dwivedi et al., 2015; Liu et al., 2010; Mpazanje et al. 2013; Nelson, 2007; Nulden, 1996). However, the increasing complexity and uncertainty of ISD projects and contemporary system development processes are challenging ISD project failure scholarship and literature. There is a need to better understand ISD projects as increasingly complex undertakings and to complement existing research with approaches that attend to and unpack these complex processes. Furthermore, contemporary system development methods and practices (e.g. agile development and continuous deployment) call for a deeper and more contextual approaches to ISD project failure. The purpose of this article is therefore to review and advance knowledge about ISD project failure/success by answering the following research questions: What are the contributions and weaknesses of the extant ISD project failure/success literature? and What are potential avenues to move the ISD literature forward? These questions are among the most important for the IS discipline, for both research and practice, and call for urgent attention.

Responding to these research questions, the article makes the following contributions to the literature:

1. It provides a critical review of the literature on ISD project failure/success by examining and assessing the major perspectives of research, their focus, assumptions, contributions and weaknesses.
2. It identifies and defines the important but underexplored phenomenon of ISD project distress as a troublesome state of a project that increases the risk of failure.
3. It proposes new areas for research by applying a sensemaking framework (drawing on Weick, 1995) for examining ISD project distress and developing a corresponding research agenda.

The first part of the article reviews the literature on three major perspectives according to their underlying philosophical assumptions about ISD project failure (based on Fincham, 2002).¹ The most established theoretical stream takes a *rationalist perspective* on ISD project failure, explaining project failure as causally linked to technological, social and organizational factors (e.g. DeLone and McLean, 2003; Lyytinen and Hirschheim, 1987; Petter et al., 2008). This perspective has been criticized, however, for abstracting the context of ISD projects and failure assessment (as it generalizes across contexts; Wilson and Howcroft, 2002) and for adopting a static, deterministic and objectivist view of ISD project failure.

In response to these weaknesses, the second theoretical stream takes a *process perspective* and examines ISD project failure as a result of project flaws and social and

political organizational processes (e.g. Kim and Pan, 2006; Lam and Chua, 2005; Pan et al., 2004; Sauer, 1993). This perspective offers more insights into the idiosyncratic nature of ISD projects and their management, arguing that, for example, ISD projects fail due to inappropriate responses to organizational problems and flaws. While this perspective contributes to better understanding of the organizational processes and flaws leading to ISD project failure, it objectifies process flaws and focuses on failure as a clearly defined outcome of such processes.

The third theoretical stream departs from the first two by taking *narrative and social constructivist perspectives*. Studies within this stream propose that ISD project failure is too subjectively ascertained (e.g. Bartis and Mitev, 2008; Fincham, 2002; Fitzgerald and Russo, 2005; Sarosa and Zowghi, 2005; Savolainen et al., 2012). However, this research perspective has been criticized for lack of generalizability and actionable advice (Fincham, 2002).

Across the three theoretical perspectives, two common weaknesses have to be pointed out. First, the research outcomes are not sufficiently actionable in ISD project management (e.g. Doherty et al., 2012; Ramiller and Pentland, 2009). As Yeo (2002) warned, ‘the gap between theory and practice in IS studies and in particular failure studies’ persists (p. 246). Second, the framing of ‘project failure’ has been largely outcome-focused, either examining factors impacting project failure or social/political processes leading to project failure. The body of research has thus become dominated by studies framing ISD project failure as an end state (i.e. the final outcome). These shortcomings serve as the impetus to examine ISD projects before they become ‘failed projects’.

Based on the literature review and a critical assessment of the three theoretical perspectives on ISD project failure, the second part of the article focuses on ‘ISD project distress’, a troublesome project condition that threatens a project’s successful completion (a detailed concept development is provided in section ‘Towards ISD project distress’). In such a condition, an ISD project experiences major disturbances that require heedful responses and intervention. By drawing attention to ISD project distress, we emphasize the situated and dynamic nature of ISD projects and the conditions that increase the risk of failure (see also, Keil et al., 2013). This focus represents a major expansion of ISD project failure research to project distress that opens up new questions regarding understanding and acting upon such project condition in an agile and timely manner. With this motivation, we propose a sensemaking theoretical framework (drawing from Weick, 1995) to examine the situated, fluid and dynamic nature of ISD project distress. This framework allows us to explore avenues for shifting the focus of ISD project failure research towards ISD project distress and propose a research agenda. The article concludes by summarizing the arguments for reorienting research to ISD project distress and discussing the expected contributions to the IS literature and practice.

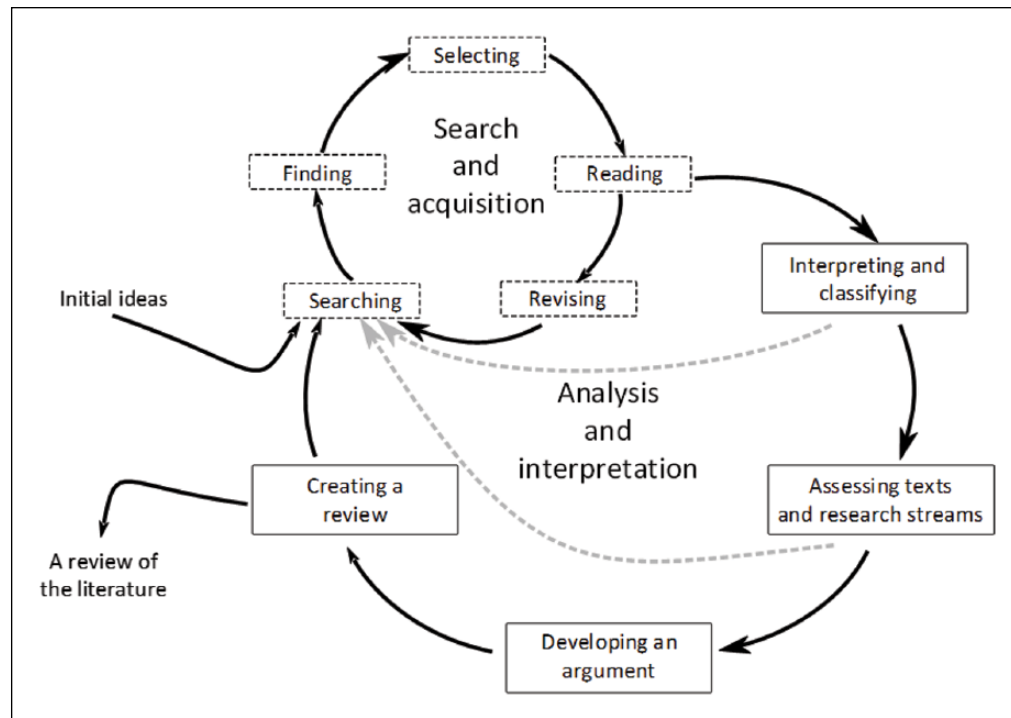


Figure 1. Literature review process involving two hermeneutic circles (adapted from Boell and Cecez-Kecmanovic, 2014: 264).

Literature review method

We conducted a review of the IS and project management literature on ISD project failure by adopting the hermeneutic framework for conducting literature reviews (Boell and Cecez-Kecmanovic, 2014). Drawing from the philosophical tenets of hermeneutics (e.g. Barrett et al., 2011; Gadamer, 1976; Ricoeur, 1981), the hermeneutic literature review approach does not assume that the meaning is given (hidden) in the text and then ‘recovered’ by a reader, but rather assumes that the meaning emerges through an encounter (a dialogue) between a text and a reader. In other words, the literature review process is an inherently interpretive process through which researchers expand and deepen their understanding of relevant literature. This further implies that the relevant literature on a particular topic does not exist irrespective of a researcher. Rather, the body of relevant literature is constructed through a hermeneutic process seeking *verstehen*, that is, interpretive understanding. Importantly, this process is intrinsically iterative as the researcher interprets a text (a paper) from her or his pre-understanding of a literature and then develops better understanding of the literature based on each interpreted text, and so on. This is called a hermeneutic route to understanding or the ‘hermeneutic circle’ (Gadamer, 1976).

We used the hermeneutic literature review approach shown in Figure 1. The first circle, ‘literature search and acquisition’, consisted of searching, finding, selecting and (surface) reading of relevant papers, followed by revised searches (the first hermeneutic circle, upper half of Figure 1).

This circle was embedded in a second, wider hermeneutic circle of ‘analysis and interpretation’, which in an iterative manner involved deep reading, interpreting and classifying papers and identifying research streams, followed by assessments of their contribution (the second hermeneutic circle, lower half of Figure 1). Our review went through these two circles several times leading to an argument development and creation of the literature review. By going through the embedded hermeneutic circles as shown in Figure 1, we developed a comprehensive understanding of research on ISD project failure and recursively created the body of relevant literature. We also identified the major research streams, assessed their contributions to knowledge and weaknesses as well as overall shortcomings of the literature.

We started the literature review based on our initial knowledge about ISD project failure literature and also searched for review papers and highly cited empirical papers in this domain. Based on these papers, we developed a set of keywords (i.e. ISD project failure or success, ISD project evaluation or assessment, ISD projects and problematic situations) to search various databases (EBSCO, ScienceDirect, Scopus, Web of Science and Google Scholar). Initial searches led to new references and in turn additional keywords such as project turnaround, de/escalation and process flaws. Repeated searches produced additional sources from which we selected a further list of relevant papers worth deeper reading. This iterative process of searching, reading and assessment was an effective way to gain an overview of past and current research and

discourse in the ISD project failure literature. From this starting point, we progressed by deeper reading of the acquired papers, followed by further searching. We then used the snowballing and citation pearl growing techniques (Boell and Cecez-Kecmanovic, 2014) to find additional relevant sources. Through such an iterative process, we identified and selected a set of 80 relevant sources (papers, chapters and books) worth further deep reading and exploration, thus entering the wider hermeneutic circle – analysis and interpretation.

We proceeded with in-depth reading of the initial set of sources and identified and classified distinct research topics, approaches and research streams. We examined, compared and debated various streams and their assumptions regarding the existence of project failure (ontological assumptions) and ways of knowing about such failure (epistemological assumptions). We were also able to develop broader understanding of the ISD project failure literature that allowed us to identify different research streams (the rationalist, process and narrative streams which accorded with Fincham's, 2002, classification) and classify individual papers. Such classification of papers revealed topics and issues specific to the research streams leading us back to searching (indicated with the dotted line from 'interpreting and classifying' to 'searching' in Figure 1). The new searches helped us find additional sources contributing to each stream and also, and importantly, enabled us to test comprehensibility of the body of literature and its classification. We then proceeded with hermeneutic analysis by developing deeper understanding and assessing contribution to knowledge of each stream as well as of individual sources. As part of this process, we also checked these assessments by conducting additional specific searches (indicated with the dotted line from 'assessing text and research streams' to 'searching' in Figure 1).

Through the above hermeneutic literature review process, we developed a substantial body of ISD project failure literature comprising in total 321 relevant sources – papers, chapters and books. The resulting literature review classified the sources into research streams, assessed contribution to knowledge by each stream and developed an argument regarding this body of literature. The remainder of this article is based on this review of ISD project failure literature, including our critical assessment of the nature of knowledge developed and the relevance to practice. We note that not all 321 sources are explicitly mentioned in our review below. While we examined all the sources to develop the descriptions of individual streams in our analysis and discussion, we quote papers that exemplify research in each research stream.

Research perspectives on ISD project failure

Several different definitions of ISD project failure have been offered in the literature (Doherty et al., 2012; Fincham,

2002; Keil and Montealegre, 2000; Lee and Xia, 2005; Lyytinen and Hirschheim, 1987; Robinson, 1994; Sauer, 1993; Thomas and Fernandez, 2008). Importantly, there is no clear consensus on the definition of ISD project failure. In general, the definitions of failure in the literature are typically based on the (different) perceptions or opinions of the people surveyed. Definitions include aspects such as going over time and budget, not meeting users' specifications, expectations or objectives, and project termination or abandonment. The lack of consensus among researchers regarding the definition of failure (and success) presents an issue in terms of comparing and interpreting empirical studies (Dalcher and Drevin, 2003; Ika, 2009; Wilson and Howcroft, 2002). Furthermore, there is often a lack of a clear distinction between project failure and success of ISD projects (Hart and Warne, 2007; Lyytinen and Hirschheim, 1987). For example, Nelson (2005) presents a case of 'failed success'; Bartis and Mitev (2008) investigate the case of a failed ISD project that was presented as a success; and Cecez-Kecmanovic et al. (2014) examine the simultaneous enactment of an ISD project as both success and failure. Given this ambiguity of ISD project failure in practice, it is not surprising that researchers have not only defined ISD project failure very differently, but have approached research on ISD project failure from very different perspectives.

We identified three major perspectives on ISD project failure in the literature, which we classified according to their underlying philosophical (ontological and epistemological) assumptions (see also Fincham, 2002; Mukherjee, 2008; Orlikowski and Baroudi, 1991; Thompson, 2011). Our classification is consistent with Fincham's (2002) and distinguishes between the rationalist perspective, the process perspective, and the narrative and social constructivist perspectives.

Rationalist perspective on ISD project failure

The rationalist perspective considers an ISD project to be a single entity, a 'black box' described by a set of attributes. ISD project failure (or success) is then an objective, measurable, and clearly determinate end state, existing independently of an observer (realist ontology). Furthermore, research within the rationalist perspective assumes that ISD project failure is causally linked to or determined by various factors, such as managerial, organizational and technical factors. Many research studies thus seek to identify these factors (e.g. Joslin and Müller, 2015; Kawamura and Takano, 2014; McManus and Wood-Harper, 2007). Furthermore, research studies propose and test general models to establish causal relationships between the factors and project failure. Ultimately, the models aim to explain and predict project failure. By identifying a range of behavioural, managerial, organizational and technical factors as causes of ISD project failure, these research studies aim to assist managers in practice: by paying attention to these

factors, managers are expected to increase the chances of avoiding project failure (e.g. Lyytinen and Hirschheim, 1987; McManus and Wood-Harper, 2007; Petter et al., 2008; Yeo, 2002).

As an example of this perspective, Yeo (2002) classified ISD projects in three categories based on their final outcome (based on the Standish Group's 'Chaos Report' in 1995). According to Yeo's classification, 'successful projects' are those that have produced and implemented IS that fulfil all planned functions and features. 'Challenged projects' are those ISD projects that have not progressed according to the business plan and have less functionality than originally planned. 'Impaired projects' are those ISD projects that have been cancelled or abandoned during the development process. Overall, 83.8% of all ISD projects are classified as either challenged or impaired projects. According to the analysis, the top five factors causing failure are 'underestimated timeline', 'weak definitions of requirements and scope', 'inadequate project risk analysis', 'incorrect assumptions regarding risk analysis' and 'ambiguous business needs and unclear vision' (Yeo, 2002: 245). McManus and Wood-Harper (2007) identified three classes of key reasons for a project cancellation: 'business reasons', 'management reasons' and 'technical reasons'. Kawamura and Takano (2014) investigated the reasons for the high failure rate. The authors state that they aim to reduce project failure rates by identifying the factors relevant for project outcomes. Based on their conclusions from a multivariate analysis, they suggested seven factors that significantly affect the ultimate project performance and its outcomes: *'company's skill and requirement, project planning, detailed planning and product quality in each phase, project monitoring and control, change requirement management, skill and teamwork of project members and schedule progress in each phase'* (p. 689). In particular, they emphasized the importance of project planning.

The rationalist research perspective has been dominant in the IS literature. This literature has contributed a significant body of knowledge that has identified the major factors causally related to ISD project failure, generalized across contexts and time (as summarized in Table 1).

However, the rationalist perspective has been criticized for a range of reasons. Particularly, for a lack of a deeper understanding of complex ISD phenomena and for the use of simple, linear cause-and-effect models to explain ISD project failures (Fincham, 2002; Sauer, 1993). This criticism goes to the heart of the rationalist approach as it is central to the approach to abstract the specific contextual details and intricacies of ISD projects in particular organizational, cultural, social and historical contexts in order to 'generalize' across contexts. Furthermore, Fincham (2002) also critiqued the view of the ISD project as a black box with certain properties, and failure and success as 'objective and polarized states' that can be unambiguously measured (p. 3).

Specifically, the idea of providing the list of factors causally related to ISD failure in the rationalist perspective has been criticized. Such lists of factors, 'no matter how lengthy and inclusive, do not provide coherent explanations of why failure occurs, and rarely explain failure across different organizations' (Fincham, 2002: 4; see also Kraemer and King, 1990). Furthermore, lists of a priori factors impacting ISD project failure are not of much help to project managers when faced with project problems and crisis in the moment. Furthermore, numerous sources (e.g. Chih and Zwikael, 2015; Samset, 2009; Standish Group, 2015; Zwikael and Smyrk, 2012) raise the possibility of the occurrence of project failure even when typical measures of success, such as the requirements of the 'iron triangle' (i.e. being on time, on budget and according to specifications), are met. For example, the Los Angeles Metro project fulfilled these typical measures of success, but was a failure nonetheless by all accounts (Shenhar and Dvir, 2007). What is missing in the rationalist perspective of ISD project failure research is an understanding of projects as socio-technical processes that unfold over time in a non-determinist manner. The 'process perspective' aims to address these shortcomings (Sauer, 1993).

Process perspective on ISD project failure

The process perspective understands ISD projects to be emergent, dynamic, political and social processes. The failure (and success) of ISD projects are seen as resulting from both internal project issues and broader organizational processes. A prominent model within this perspective, as proposed by Sauer (1993, 1999), focuses on organizational processes and the interaction between the IS, the IS project organization and the IS supporters (owners). This interaction is prone to various flaws that may snowball if not identified and dealt with promptly. When ISD project supporters feel that an IS is not serving their needs and interests, they may terminate the project, using project flaws as justification. Sauer's (1993) process model makes a distinction between 'flaws' and 'failures'. Flaws are objective internal project problems, often technical, while failures result from decision-making and organizational processes, and tend to be 'politically and socially defined' (Fincham, 2002; Sauer, 1993). Sauer's (1993) model therefore takes an objective approach to the analysis of project 'flaws' and their effects on organizational processes and ISD project failure. At the same time, the outcome of 'failure' also depends on interpretations made in a social context. While Sauer's model takes into account the subjective and political aspects, it still objectivizes the project's state of failure and considers this state irreversible (Fincham, 2002).

According to the process perspective, a 'problematic' ISD project may suffer from a combination of issues. Issues identified in the literature include the following: the misinterpretation of process issues as technology issues (Kim

Table 1. Three perspectives on ISD project failure (and success).

Perspective	Conceptualization of ISD project failure (and success)	Purpose of research	Level of analysis	Methodological focus	Major claims and findings	Weaknesses and underexplored aspects
Rationalist perspective	- Objective and polarized end states: either failure or success (not both). - Outcomes caused by technological, organizational and social factors.	- Explain causal relationship between failure/success and various factors. - Predict and prevent ISD project failure.	ISD projects, stakeholders and organizations.	Identify major factors causally related to ISD project failures, generalized across contexts and time.	A set of behavioural, technical, organizational and managerial factors shown to lead to ISD project failure/success.	- Generic factors determining failure/success independent of context are not seen as useful in practice. - Considers failure/success as definite end states that exist objectively (independent of the observer).
Process perspective	Exemplary papers: Lyttinen and Hirschheim (1987), DeLone and McLean (1992, 2003), McManus and Wood-Harper (2007), Petter et al. (2008, 2013), Kawamura and Takano (2014) and Joslin and Müller (2015) - An emergent, dynamic and unpredictable outcome of socio-technical interaction. - Socially and politically defined outcome of organizational processes and flaws.	- Identify ISD project flaws. - Avoid failure by making adjustments in ISD projects, based on timely detection of flaws.	Socio-technical, organizational and political processes.	Socio-technical interaction often investigated through case studies.	Ignorance of project flaws and absence of proper reaction towards them during ISD projects lead to failure.	- Objectifies ISD project flaws and their effect on organizational decision-making. - ISD project failure is seen as irreversible.
Narrative and social constructivist perspectives	Exemplary papers: Robey and Newman (1996), Lam and Chua (2005), Kim and Pan (2006), Pan et al. (2008), Sausser et al. (2009) and Mohanarajah and Jabar (2015) - Failures/successes are narrated and socially constructed. - Interpretive flexibility of ISD projects.	- Understand how ISD project failure is narrated and constructed through interpretation by relevant social groups. - Learn from failure.	Organizational and socio-political processes, symbolic action, themes, plots and stories.	- Narratives, rhetoric and persuasion. - Interpretation and meaning-making by relevant social groups.	- ISD project failures are narrated and constructed through discourses. - There are reflexive relationships between narratives of failure and narratives of success.	- No generalizable and actionable knowledge. - Focuses on social, discursive and political aspects at the expense of technological aspects; considers human agency but not technological agency.
	Exemplary papers: Bartis and Mitev (2008), Doherty et al. (2012), Fincham (2002), Fitzgerald and Russo (2005), Fortune and Peters (2005), Havermans et al. (2015) and Lim et al. (2011)					

and Pan, 2006); project recommencement within the same larger problematic pattern (Robey and Newman, 1996); strategic misdirection of a project (Pan et al., 2004); mismatch between requirements and actions (Lam and Chua, 2005); failure to recognize problems (Pan et al., 2004); a weak configuration fit between existing business processes and the ISD project (Sauer et al., 1997); escalation of commitment and ongoing dedication of resources to a failing course of action (Pan et al., 2004); misunderstandings of failure and flaws (Sauer, 1993); and the misfit between project characteristics and the project management approach (Sauer et al., 2009).

As an example of this perspective, Robey and Newman (1996) suggest that problematic situations and project failures tend to result from a repeated problematic pattern of actions in relation to ISD processes (instead of developing a new pattern of actions). According to Robey and Newman (1996), it is the inability to detect and correct the pattern that might be the 'root cause' of ISD failure. They believe that attention should be given to identifying patterns and reflecting on 'how to reverse repeating patterns of failure' (p. 60). The process perspective suggests that adjustments in ISD projects are required when the current procedures become flawed, potentially leading to project failure. For this, objectively existing problems and flaws must be identified. In Pan et al.'s (2004) case study of a troubled project, management initially underestimated how influential the roles of external suppliers on the project were and did not inform the higher management of any critical problem arising. The troubled project was successfully turned around by a timely intervention and the implementation of a 'de-escalation strategy' (p. 261).

Compared to the rationalist perspective, the process perspective offers important insights into the complexity of ISD project failures. It provides more clarity on socio-technical interactions and the organizational and political dimensions of judgements on ISD project failure (as summarized in Table 1). However, the assumed objectivity of project flaws, problematic patterns of action and the ways that they may affect the 'failure' verdict still preserve some rationalist assumptions (Fincham, 2002) that limit the explanatory potential of this perspective.

Narrative and social constructivist perspectives on ISD project failure

The third research stream includes two related and overlapping perspectives, namely, the narrative and social constructivist perspectives. They both emphasize the subjective, discursive, interpretive and political nature of the assessments of ISD projects. These perspectives focus on everyday interactions between different stakeholders or social groups and the ways in which they narrate and discursively construct their reality, including the assessment of ISD projects as failures (or successes). They hence provide a more

radical departure from the rationalist perspective compared to the process perspective.

Narrative perspectives have been developed in IS research as part of a broader non-positivist research movement (Walsham, 1993). The narrative perspective aims to understand how actors narrate and make sense of projects and how they attribute failure within their interpretive schemes (rather than to understand the causes of ISD project failure). The narrative perspective has some similarity with the process perspective in that it considers the interaction between social/human and technological/mechanical processes. However, it avoids the view of ISD project failure as an objective outcome, found in both the rationalist and process perspectives. Instead, the narrative perspective considers the subjective aspects of meaning-making and the emergence of project narratives, including plots, and themes of failure and success.

As an example of this perspective, Fincham (2002) presented two case studies of financial IS development by focusing on tensions emerging between narratives that affirmed a new social order and narratives that re-affirmed the old social order. In both cases, discourses alternated between 'strategic' and 'pragmatic' narratives, resulting in the projects being labelled differently, as success or failure, at different times. As Fincham (2002) concluded,

[Success and failure] were a complex double act rather than polarized outcomes. They were mutually defining constructs used by organizational actors in response to the circumstances they confronted. In this respect neither of the case studies were simple either/or cases. Both contained episodes of success and failure that were part of an unfolding series of events, the main difference being the particular stage in the story in which the observer enters. (p. 13)

According to the narrative perspective, project narratives are fluid, dynamic and uncertain: changing social and power relations construct and alter the meanings of success and failure over time. In this sense, 'narratives are both about and become the change process' itself (Buchanan and Dawson, 2007: 669).

The narrative perspective adopts a 'methodological position through which to engage not with a presumed neutral "real" world, but with the complex nuances of the "lived" world' (Rhodes and Brown, 2005: 180). This perspective adds another layer of discursive actions to the analysis of ISD processes, adding questions about the 'why' (e.g. political reasons) to the 'how' and 'when' (the questions of the process perspective). This perspective demonstrates the role of narrative strategies, labelling, dramatic plots and discursive actions in the assessment of ISD projects. It focuses on the reflexive and mutually defining relationship between narratives of failure and narratives of success (Fincham, 2002).

Closely related to and overlapping with the narrative perspective, the social constructivist perspective assumes

‘interpretive flexibility’ of ISD projects, emphasizing the contingent nature of project assessment. In this perspective, there is no objective definition of ISD project failure as it exists only as interpretation or construction by relevant social groups (Chen et al., 2009; Dougherty et al., 1998; Wilson and Howcroft, 2002). This perspective demonstrates how failure is constructed in the context of different interpretive schemes, which in turn are shaped by political and social groups and practices.

As an example of this perspective, in a case study by Bartis and Mitev (2008), the project committee reported their ISD project as a success, while the actual users found the IS inappropriate for their intended use and considered the ISD project a failure. The authors demonstrated how the construction of the powerful social group (project committee) prevailed, concealing the project’s failure. This case illustrates how different relevant social groups or stakeholders attribute different meanings to an ISD project and construct different assessments as a result of underlying organizational dynamics, power relations and cultural norms. Havermans et al. (2015) drew attention to the use of language in framing the issues and stories around projects as a vital tool in shaping projects and reaching problem resolution. They analysed project managers’ narratives of emerging problems to explore the storylines they construct as well as the way that they affect the collaborative efforts of different groups and the ways they can lead to project failure or success. They concluded that the leaders’ narratives addressing emergent project problems had significant impact on the way in which the project team approached and resolved them.

The narrative and social constructivist perspectives have contributed a deeper understanding of how the narratives and processes of meaning-making and interpretation in specific social and political contexts construct the failure/success of an ISD project. These perspectives intend to produce situated and context-specific knowledge. However, these perspectives have been criticized for contributing idiosyncratic findings, for failing to produce generalizable and actionable knowledge, and for overemphasizing social issues over technological/material issues (Doherty et al., 2012).

In addition to the three major perspectives on ISD project failure discussed above, we found more recently published research attempting to extend the social constructivist perspective (Cecez-Kecmanovic et al., 2014; Mpazanje et al., 2013; Rao and De’, 2015). While they see failures of ISD projects as constructed, they claim that failures are not only socially constructed but also *performed* by non-social actors. This emergent perspective proposes that by continuously reconfiguring the relationships between heterogeneous (human and non-human) actors in ISD projects, failure is performed as a temporal effect. Consequently, failure is not an end state but rather a dynamic effect of the complex relationships of both social and technological actors. This approach is promising in advancing ISD project failure

research, but it is in an early stage with a few examples (hence not considered as a separate, major perspective here).

Critical assessment

Questions of failures (and successes) of ISD projects are among the key topics in the IS literature. They continue to attract researchers of different persuasion. The three established research perspectives have produced vast and significant knowledge, highly valued in the IS discipline. Table 1 summarizes these perspectives, providing for each the conceptualization of ISD project failure/success, the purpose of research, the level of analysis, the methodological foci, the major claims and findings, as well as the weaknesses and underexplored aspects (as discussed above).

High failure rates of ISD projects in practice, however, persist calling for further research and expansion of research beyond existing approaches (Doherty et al., 2012; Rao and De’, 2015). In particular, the increasing uncertainty and complexity of ISD projects and their assessments led some IS researchers to question the ways in which ISD project failures have been approached, framed and studied (Bussen and Myers, 1997; Cecez-Kecmanovic et al., 2014; Doherty et al., 2012).

In our assessment, one of the key issues in the established research perspectives is the focus on ISD project failure as a distinct end state. This is particularly prominent in the rationalist perspective in which project failure and success are seen as fixed, objective and measurable end states that exist independently of the observer or assessor. The process perspective has extended the focus to organizational decision-making processes and project flaws, but still focuses on the effect of project flaws on the end state which, in the case of failure, is considered irreversible. The narrative and social constructivist perspectives go further in examining how project success and failure are narrated, interpreted and constructed by the relevant social groups. While research in these perspectives examines the narratives and processes of failure construction, their focus remains on the ultimate end state of ISD project failure. That is, while the existing literature placed much emphasis on the final assessment or end state of ISD projects as failures (or successes), it did not show sufficient interest in studying problems and problematic situations during ISD projects, the way that they arise and how they are addressed.

In addition to the reviewed literature on ISD project failure, there were studies focusing on the factors that contribute to continued commitment of resources to a project for which decision-makers receive ‘negative information’ about project performance (e.g. time or budget overruns). This phenomenon is commonly termed ‘escalation of commitment’ (Keil, 1995; Mähring and Keil, 2008; Newman and Sabherwal, 1996). Related studies examined factors impacting ‘de-escalation of commitment’ that could include

project redirection or abandonment (Keil and Robey, 1999; Montealegre and Keil, 2000; Pan et al., 2006; Pan and Pan, 2011).

Escalation studies differ in terms of seeing and theorizing project continuation decisions after receiving negative information as *irrational* or *rational* (Desai and Chulkov, 2009). The irrational escalation models identify factors or forces (such as self-justification by responsible managers, preserving consistency in decision-making, negative framing of project termination due to large sunk funds, and organizational inertia) that contribute to decisions to continue resourcing a 'failing' ISD project. To address irrational escalation, de-escalation techniques are proposed including improved governance and regular monitoring and evaluation of projects so that the failing projects are terminated or redirected early (Keil and Robey, 1999; Montealegre and Keil, 2000; Pan et al., 2006). Other studies, however, found that escalation of commitment is not necessarily irrational, as managers and organizations assign value to a project despite, for instance, its running over time or over budget (Keil and Mann, 1997; Keil et al., 2000). Furthermore, escalation research has been critiqued for interpreting managers' commitment to high risk and innovative ISD projects as irrational escalation (Chulkov and Desai, 2005, 2008).

The underlying assumption in these escalation studies is that the so-called failing projects can be known and are detectable in advance, based on negative information from insiders or assessment by auditors. This implies a deterministic view of project failure that has been critiqued in the literature on success/failure of ISD projects that emphasizes uncertainty and complexity of both ISD projects and their evaluation processes (see e.g. Bussen and Myers, 1997; Cecez-Kecmanovic et al., 2014; Doherty et al., 2012). Furthermore, the assumption that managers would get and act (rationally) upon receiving negative information about a project, which will lead to de-escalation, is quite optimistic because managers are often shielded from receiving the 'bad news' (Snow et al., 2007; Thompson et al., 2007; Keil et al., 2007, 2014; Iacovou et al., 2009) and also because they avoid accepting responsibility for failure or taking action that would damage their reputation (Koster and Lambooj, 2018; Lee et al., 2014).

While these escalation studies contributed to better understanding of the behaviour of decision-makers (managers who allocate resources to ISD projects) and the factors that contribute to their commitment to 'failing projects', they did not explain the actual process of escalation or the transition to de-escalation (Mähring and Keil, 2008; Pan and Pan, 2011). This is addressed by so-called process models of escalation. For instance, Mähring and Keil's (2008) process model involves three phases – *drift*, *unsuccessful incremental adaptation* and *rationalized continuation* – with project framing as antecedent condition and de-escalation initiation as the outcome. The process model

of escalation draws attention to problems that start often with project framing and continue during project execution. Project escalation occurs, Mähring and Keil (2008) explain, 'because actors that exercise influence over an IT project repeatedly respond to indications of problems in a way that does not match the nature and severity of the actual underlying problems (thus failing to resolve problems)' (p. 265). Similarly, Pan et al.'s (2009) process model describes antecedent conditions (history from past projects), sequences of events (including a project level and work level events), critical incidents and outcomes, over the course of an ISD project. Importantly, the model identifies the 'recursive interactions between [ISD] project, organizational work activities, and their contexts' (Pan et al., 2009: 1), using the approach-avoidance theory (Keil et al., 2000). The model enables a 'deeper understanding of the ambiguity and dilemma that decision makers face during project escalations and de-escalations' (Pan et al., 2009: 76).

In summary, process models of escalation add important knowledge to ISD failure literature by explicitly addressing emerging problems during ISD projects and also identifying different mechanisms (processes) by which unsuccessful coping with the emerging problems fuel the escalation process (Keil et al., 2014; Mähring and Keil, 2008; Pan et al., 2009). Although process models of escalation remain largely focused on decision-makers and decision-making processes leading to continuation of investment or transitioning to de-escalation (Drummond and Hodgson, 2011; Koster and Lambooj, 2018), they indicate a promising direction to explore further critical problems emerging during ISD projects in their broader organizational context and ways of dealing with them.

Overall, based on the assessment of the ISD failure literature and also inspired by the process models of escalation literature, we seek a theoretical (re)framing of ISD project failure that allows for a deeper and more nuanced understanding of how problematic situations emerge and how actors in ISD projects (and beyond) identify and deal with (or should deal with) those situations. An improved understanding of an 'emerging failure' could help to better interpret why some projects fail and others succeed, even given similar technological, social and organizational factors. Moreover, better understanding of problematic situations and ways of dealing with them in ISD project practices could help project managers (and other actors involved in ISD projects) in identifying, dealing with, and possibly 'rescuing', projects that experience serious problems.

In the remainder of the article, based on the review and critique provided, we therefore develop a reframing of the key research problem as one of 'ISD project distress'. As we argue below, this reframing will allow us to better focus on the situated and dynamic conditions of ISD projects and the troublesome situations that threaten successful project completion. Such reframing will open up a new avenue for ISD failure research that promises to address the current

weaknesses in the literature and create relevant and applicable knowledge. The purpose of articulating the concept of 'ISD project distress' is to move ISD research forward through developing an alternative problem framing, for which we are proposing a research agenda.

Conceptualization of ISD project distress

Towards ISD project distress

ISD project distress, as a coherent concept, has not been explicitly conceptualized and studied in the literature. We propose the term 'distress' by drawing from medical terminology.² Distress in the original medical sense refers to a harmful human condition 'that tends to disturb the balance of body and mind and promotes ill health [and a] condition of being in need of immediate assistance' (Free Dictionary, 2016). Distress is also a 'negative response to stressors' (Nelson and Simmons, 2003: 266).

We define ISD project distress as follows: An ISD project is in a state of distress when it experiences critical problems at any stage of the development process. A state of distress denotes harmful conditions in an ISD project that cannot be routinely resolved and that require heedful attention and agile responses. ISD project distress is a dynamic and fluid constellation of critical problems that typically may not be easily identified, understood and addressed.

The opposite of a 'distressed' ISD project is a 'smoothly running' ISD project. However, this is not a clear either/or distinction, and we note that stress (periods of high tension or activity) does not necessarily have to become distress but might be handled without negative impacts on actors or the project, if promptly and effectively acted upon.

Several problems and questions pertaining to what we call ISD project distress have been discussed in passing in the ISD project failure literature. For example, Fortune and Peters (2005) reported several ongoing problems related to project performance that persisted throughout an ISD project. Standish Group (2009) reported on different problematic situations in ISD projects. Kautz et al. (2007) identified 'persistent problems' during a mobile application development. Kappelman et al. (2006), in a post-mortem examination of failed IT projects, talked about 'early warning signs' as significant symptoms throughout projects well before failure occurs. Yet, while some important problems during ISD projects have been mentioned in the literature, there has been no integrating concept for such problems.

Research on 'early warning signals' (Kappelman et al., 2006; Nikander and Eloranta, 2001) found that early warning signals of future project problems may include the following: personnel or project issues; misunderstandings due to cultural and linguistic differences; conflicting or

inadequate information; miscommunication; or simply a 'gut feeling' that there are problems beneath the surface. These signals often become stronger and more specific over time and can help project managers to identify and predict problems and take timely actions.

However, beyond identification of early warning signs, there are unanswered questions of diagnosing and understanding project distress, of seeking an appropriate intervention (response) to distress and mobilization of all relevant actors in this process. It is important to recognize that project distress is a critical state that ultimately might lead to project failure, but that such outcome is not inevitable. In other words, if distressed condition of an ISD project is not dealt with and the project situation deteriorates the likelihood of the project outcomes being labelled a 'failure' increases. However, if a distressed condition is identified and diagnosed early enough and effective corrective actions are promptly taken, a distressed ISD project is likely to become a smooth running with expectations of 'successful completion'. A distressed ISD project can be dealt with and transformed – however, very few studies have looked at how this transformation can be achieved. Exceptions are escalation process studies (Keil et al., 2014; Mähring and Keil, 2008; Pan et al., 2009) that show how decision-making processes deal with project problems and lead to escalation or de-escalation of commitment. Due to the lack of focus on ISD project distress in the IS literature, however, the distressed project conditions and states are neither systematically studied nor well understood and consequently a range of potential interventions to remedy distressed situations are not explored. What is needed is the conceptual development of, and research focus on, ISD project distress.

A multilevel sensemaking framework for ISD project distress

We use a multilevel sensemaking approach to account for the fact that ISD projects are complex human, social, organizational and technological phenomena. During the course of ISD projects, there are various emerging and ongoing problems and troubles originating in business/legal environments, organizing and managing processes, within a project or between projects, teams and team members (Kautz et al., 2007). Importantly,

in real-world practice, problems do not present themselves as givens. They must be constructed from the materials of problematic situations, which are puzzling, troubling and uncertain. In order to convert a problematic situation to a problem, a practitioner must do a certain kind of work. [S/he] must make sense of an uncertain situation that initially makes no sense. (Schön, 1983: 39–40)

In the practices of ISD projects, we need to account for how different actors 'make sense' of issues and troubles, articulate problems and find ways to solve them.

To develop the concept of ISD project distress, we build on the notion of sensemaking as the ‘process through which people work to understand issues or events that are novel, ambiguous, confusing, or in some other way violate expectations’ (Maitlis and Christianson, 2014: 57) and through which they engage in meaning-making, interpretation, articulation and action (Weick, 1995). While sensemaking involves interpretation, it goes beyond interpretation as making sense of problematic situations implies the active authoring of events and constructing the situations and actions (Weick, 1995; Weick et al., 2005). While sensemaking has been variously conceptualized in the literature, we draw from Weick’s (1995) influential work on sensemaking in organizations as it is highly relevant to the problem space of ISD projects in organizational contexts. Most importantly, Weick views sensemaking as a central activity of an organization, at the core of organizing. His holistic concept of sensemaking incorporates individual, collective and social sensemaking and is also both retrospective and prospective (Maitlis and Christianson, 2014). In this section, we first describe Weick’s (1995) generic four-level sensemaking model and then adapt this model for conceptualizing ISD project distress.

Weick’s sensemaking model. Weick’s (1995) sensemaking model allows us to propose a theoretical framework within which the nature of ISD project distress can be better articulated, understood and examined. In particular, his four-level sensemaking model, based on the theory of the semiotic self (Wiley, 1988, 1994), distinguishes sensemaking processes at four levels (Ceccez-Kecmanovic, 2004):

1. The *intra-subjective sensemaking* level refers to an individual (self, actor or subject) who makes sense of the events and actions around him or her. This sensemaking involves individual experiences, thoughts, beliefs, feelings, desires, knowledge, skills and so on. While intra-subjective sensemaking is not an isolated process, at this level, ‘meaning is within the self, and the subject is, by definition, fully present’ (Wiley, 1994: 154).
2. The *inter-subjective sensemaking* level is collective in nature and involves people in social interaction. At this level, people are making sense collectively, while co-creating meanings and producing new sense through discussion, sharing of experiences and cooperative interpretation. The meanings emerge when ‘individual thoughts, feelings and intentions are merged or synthesized into conversations etc., which consists of an intersubject, or joined subject or merged subject’ (Weick, 1995: 71).
3. The *generic-subjective sensemaking* level or social structure level involves generic meaning shared by members of an organization. This sensemaking implies structures, norms and rules that determine

and limit meanings and foster the creation and maintenance of generic meanings shared by organizational members, while they are not necessarily involved in their creation.

4. The *extra-subjective sensemaking* level or the level of culture denotes a symbolic reality, conceptualized as an abstract idealized framework derived from prior interactions and common experiences and forms of life (Wiley, 1994). These culturally communicated and embodied meanings underpin all other sensemaking levels.

These sensemaking levels are only conceptually distinguished. Sensemaking processes at all levels are mutually intertwined, continually influencing each other (Ceccez-Kecmanovic, 2004). For instance, individual intra-subjective sensemaking is inevitably embedded in a collective sensemaking, and both are in turn grounded in generic and cultural meanings. Conceptual distinction among the levels, however, provides a useful analytical lens for understanding the complex dynamics of sensemaking in a variety of organizational contexts (Weick, 1995). We draw from this generic sensemaking model and adapt it to the context of ISD projects. Our aim is to develop a sensemaking theoretical framework for understanding and examining ISD project distress.

A sensemaking framework for ISD project distress. An ISD project can be seen as a particular type of ongoing sensemaking processes within a broader context of organizational sensemaking. That is, ISD projects are a particular form of organizing, involving people and technologies, development practices, methodologies, governance structures, policies, norms and rules, delivery plans and deliverables within a larger organizational context that includes many other organizational processes and practices. An ISD project in an organization can thus be seen as consisting of both *intra-project* and *extra-project* sensemaking processes.

All human actors in an ISD project, including analysts, users, system architects, developers, contractors, testers and managers continually and often unconsciously make sense of project objectives, requirement specifications, technological tools and everyday activities by rendering them intelligible within their interpretive schemes. These individual sensemaking processes are part of, and integral to, collective inter-subjective sensemaking within a project, involving project team interaction, and the coordination and articulation of project objectives, activities and tasks. We call these *intra-project sensemaking* processes.

Intra-project sensemaking processes are to be distinguished from *extra-project sensemaking* processes. Extra-project sensemaking involves the structural, governance level and the cultural, symbolic level. At the structural, governance level, an ISD project is seen as a single entity spending resources to produce desired outcomes relevant for the

broader organization and the achievement of strategic goals. The cultural, symbolic level, while underpinning meaning making at all other levels, is specifically relevant for the broader perceptions of a project and for embedding its meaning and importance within an organizational context.

The intra-project and extra-project sensemaking processes overlap in *project management sensemaking* processes. On the one hand, ISD project management is concerned with running the project within budget and time constraints, the planning and execution of specific activities by the project team, the availability of necessary IT infrastructure and tools, and managing the project team – all involving intra-project sensemaking. On the other hand, ISD project management is responsible to a governance body (often consisting of representatives of project owners and stakeholders) for the successful running and completion of a project, within policies and rules established by the governance structure and broader values and norms implied by organizational culture, that is, project management is also involved in extra-project sensemaking. Project manager and other actors involved in project management are hence ‘wearing two hats’ (a common saying in project management).

Processes at all three levels, the extra-project sensemaking, the intra-project sensemaking and the project management sensemaking, are all potential sources of project distress. More generally, distress originates in the inherent conflict between an organizational cultural/structural sensemaking and the individual/collective sensemaking. This conflict, according to Weick et al. (2005), is endemic to organizations. Cultural values, norms and tradition as well as organizational structure and governance (including IT governance structures) tend to preserve stability, unity and generic meanings (sensemaking at the extra-project level) that are in contrast to (and latently in conflict with) dynamic, emergent and often innovative sensemaking at the intra-project level. As a result, problematic situations and distress may arise due to cultural misfit between the IS being developed and organizational values, norms and traditions. Similarly, openness of communication and distributed/cooperative nature of decision-making in an ISD project may clash with more hierarchical organizational culture, formal communication and generic sensemaking.

Furthermore, specific to ISD project management, sensemaking processes are often conflicting pressures from organizational extra-project sensemaking and intra-project sensemaking. Demands made, for instance, by IT governance and ISD project sponsors cannot always be met due to actual constraints of resources (e.g. IT infrastructure, IT specialists, project team composition and location, and user involvement) and complexities in IS development that are not well understood outside of the project team. These conflicting pressures are potential causes of project distress. In addition, intra-project sensemaking may be prone to breakdowns of collective sensemaking and disruptive individual

sensemaking of team members that can also cause project distress. Table 2 provides examples of potential sources of project distress in extra-project sensemaking, project management and intra-project sensemaking.

Understanding the nature and sources of project distress and what the signals might be for specific sources of distress is the first step in researching ISD project distress. Identifying and understanding signs of distress will help practitioners and researchers focus on the sources of distress and appropriate actions to address them. As, typically, some problematic situations and various stressors are always present, it is important to identify those that are critical and threatening to the project. This is why empirical studies are needed to provide further insights and specific evidence about different sources and types of ISD project distress, the approaches and interventions taken to deal with them and the subsequent implications for the smooth running of a project.

The sensemaking model of ISD project distress described in Table 2 constitutes a multilevel framework that may be useful for understanding and systematic examination of the various sources, conditions and signals of ISD project distress in organizational contexts. We propose that the nature and dynamics of these sensemaking processes are of key importance for understanding, theorizing and researching ISD project distress. We use the above theoretical framework to draw from the reviewed literature and reinterpret reported problematic situations in ISD projects. That is, we select examples of problematic situations and other issues found in ISD projects failure studies and re-conceptualize them as sources of distress at different sensemaking levels in the framework. Note that this discussion is not intended to be exhaustive but is to illustrate the different nature of sources of distress, difficulties in interpreting signals of distress and implications for potential responses to distressing events and conditions. We thus illustrate how the framework could assist in exploring further and explaining deeper problematic situations and responses aimed to address them.

In the ISD literature, many problems are reported that are of concern to an individual project team member (e.g. Pan et al., 2008; Sauser et al., 2009). How such individuals understand, interpret and make sense of the project objectives, scope, methodology, organizational context and ongoing actions is essential for their constructive participation and action during the course of a project. An individual is always embedded in a wider context (project team, groups within the team, department, organization, etc.). Therefore, individual sensemaking is influenced by, and influences, collective sensemaking in multiple ways. Individuals may vary in their ability not only to perform their professional tasks but also in their capacity to be team players and share in collective articulation of project goals and activities and thus to contribute to collective knowledge and understanding of emerging project conditions and

Table 2. Sensemaking model of ISD project distress.

Sensemaking processes	Sensemaking levels	Potential sources of ISD project distress
Extra-project sensemaking processes	Cultural, symbolic sensemaking level	• Cultural misfit between the IS being developed and organizational values, norms and traditions may be the source of ISD project distress. • Differences in terms of openness of communication and distributed/cooperative nature of decision-making in an ISD project, on the one hand, and the organizational culture, on the other, may lead to project distress.
	Structural, governance sensemaking level	• ISD project distress may emerge due to dissonance between governance policies and rules (including IT governance) and those espoused by the ISD project. • Rigid IT governance structure with strict control of project phases and deliverables impose generic sensemaking that may stifle innovation in ISD projects and prevent adaptation to changing environments and user needs. • Overdependence on ISD project owner/sponsor (and provider of funding) and their views on the project progress and fulfilment of requirements may be a source of project distress. • Generic sensemaking and politics surrounding the ISD project's ongoing assessment and funds allocation may be disconnected from the actual practices and thus cause project distress. • Prescribed methodology (by IT governance) disconnected from ISD project practice may lead to project distress. • Sensemaking related to 'business-IS' strategy (mis)alignment and project portfolio management may lead to ISD project distress.
Overlapping intra- and extra-project sensemaking processes	Project management sensemaking level	• Sensemaking involved in the ISD project management within a given governance and management framework is an ongoing potential source of distress; these include issues arising from project manager's responsibility for project performance, cost control and meeting deadlines; and less than truthful reporting to the project sponsors and an IT governance body. • ISD project management sensemaking may be affected by competing demands for flexible and long-term stable IT infrastructure; lack of appropriate IT resources and tools and related expertise for specific parts of an IS design. • Sensemaking regarding conflicting demands for timely delivery within a budget and for completing system requirements and fulfilling emerging users' needs may lead to ISD project distress. • ISD project management sensemaking is affected by issues related to acquisition and retention of appropriately trained and experienced staff that may cause project distress. • Conflicting sensemaking as part of planning, allocation, monitoring and coordination of ISD project tasks within budgetary limitations and deadlines are potential sources of distress.
Intra-project sensemaking processes	Social interaction and collective sensemaking level	• Breakdowns in collective sensemaking caused by a lack of communication and coordination among members of an ISD project team may lead to a mismatch and conflict between segments of the system design and ultimately cause distress and delays. • Disconnect between the users' and the system analysts/designers' sensemaking may lead to users' resistance and project distress. • Internal friction and conflict in the ISD project team disrupt collective sensemaking that may turn into political manoeuvring that negatively affects collaboration and teamwork, leading to project distress.
	Personal sensemaking/performance level	• Personal sensemaking that is disconnected from the team collective sensemaking may lead to the disruption of an ISD project execution and thus cause its distress. • Performance appraisal based on individual output irrespective of individual contribution to the team performance may lead to individual sensemaking and performance that disregard overall project's goals and the team's collective efforts and cause project distress.

performance targets. The failure of project members to share interpretations and to participate in collective sense-making creates grounds for potential sources of distress.

Numerous studies investigating ISD project team issues have shown that these may be seen as breakdowns of sense-making at the collective sensemaking and social interaction level: the more complex the project and the bigger the team, the more diffused and breakdown-prone the collective sensemaking. For example, the lack of coordination among parts of the project and the discord between two different

teams working on different segments of a large ISD project are evident in Bartis and Mitev's (2008) case. In this case, the disconnect between sensemaking of the two teams could explain their failure to build a shared understanding about the IS they were developing and their blaming each other for being unreliable and uncooperative. Dissonant sensemaking and eventually the breakdown in communication became potent sources of project distress. Similarly, divergent sensemaking by users on the one hand and developers on the other can be a source of distress in ISD

projects (as shown by Van Akkeren, 2012). Early detection of these sources of distress would enable timely attention and response by project managers and the avoidance of problem escalation and harmful effects.

At the project management sensemaking level, the overlapping intra-project sensemaking and extra-project sensemaking may be a source of project distress when they are mutually dissonant (e.g. Iacovou et al., 2009; Pan et al., 2008; Sauer et al., 1997; Sauser et al., 2009). This happens, for instance, when ISD project managers are reluctant, or are not encouraged, to report truthfully about a project's progress and problems to project sponsors and/or the IT governing body. Project managers are responsible to the project sponsor for project performance and are typically required to report regularly to an appropriate IT governing body. Tensions may arise between expectations within and outside the project, resulting in potentially detrimental divergence of external and internal sensemaking about project performance and progress. For example, Pan et al. (2008) described the case of an inexperienced project manager who was reluctant to acknowledge existing problems to upper management and failed to report 'the truth and only the truth' to the project sponsor and an IT government body. Understanding problems leading to project distress could have remedied the situation much before it escalated into project failure.

ISD projects are typically funded within an organization's project portfolio that is designed to align with business strategy. Distress might emerge due to different strategic intentions behind an ISD project and organizational business strategy. An interesting example of strategic misalignment and the resulting distress is provided by Pan et al. (2004) who studied a failing electronic procurement project that was about to be abandoned. The decision was made to continue with the project after making governance changes to influence the attitudes and behaviours of project team members. While project distress has not been mentioned, the actors actually addressed and dealt with the major course of distress: open discussions about past mistakes and reflection on shortcomings in the earlier development process were conducted, without punishment, enabling team members to recover from 'their painful experiences' and to work 'cooperatively toward saving the failing project' (Pan et al., 2004: 265).

A common problem in organizations is the discrepancy between the understandings of a project by top-level managers on the one hand and by the core ISD project team on the other (Jones and Pfeiffer, 1973). These may include differences in norms and structures governing the project execution. For instance, the role of project manager, the business case and the reporting processes are determined by the project management methodology and the company's IT governance structure. These may be interpreted differently by a project team leading to potential conflicts and project distress.

The structural governance level of sensemaking processes aims to ensure that all projects are aligned with organizational IT governance policies and strategies. Problems thus arise when projects do not align, or are perceived to be misaligned, with the existing IT policies and strategy. These sensemaking processes are sometimes sources of project distress when they reflect a conflict between intra-project dynamic changes and extra-project sensemaking at the governance level. The governance level is inherently resistant to change while ISD projects often instigate change which sometimes causes project distress. This is exemplified by a case of agile development and management methods adopted by ISD projects that were in conflict with wider project management rules and governance structures (Ramesh et al., 2010). Such sources of distress typically remain hidden and persist without attention and intervention with potential detrimental effects on projects and organizations.

Furthermore, ISD project distress can result from a cultural misfit between the values and assumptions about the IS being developed and existing organizational values and traditions. As Heeks (2002) pointed out, 'the greater the mismatch, the greater is the risk of failure' (p. 104). Lam and Chua (2005), for example, analysed a knowledge management ISD project case and documented how organizational culture prevented the knowledge sharing central to the new IS. There was resistance to making contributions and users did not engage with the system. The authors conclude that the project would have needed a more open organizational culture. This is an example of project distress at the cultural sensemaking level that was neither recognized nor addressed before it was too late, leading to project's 'drag and slip'.

These examples of ISD projects illustrate how the recognition of ISD project distress, including its nature and sources, might provide a novel view of troubled projects and lead to timely and insightful exploration of emerging problems and their early resolution. The extant literature, however, pays insufficient attention to problematic situations during ISD projects and does not recognize the necessity to specifically focus on and deal with ISD project distress.

ISD project distress research agenda

Grounded in the sensemaking model of ISD project distress introduced above, we propose the following research agenda for studying ISD project distress. Such distress can be investigated at intra-project, extra-project and project management sensemaking levels. We use these levels to organize the research agenda.

Intra-project sources and causes of distress are related to personal and collective sensemaking within a project. They occur most often when discord exists among team members or between the project team and the users. Research questions for future research at this level include the following: How do individual team members make sense of and

negotiate project objectives, scope, structure, methodology, deliverables and deadlines? How and why do personal sensemaking processes within a team differ and what are the consequences? How do distributed and often over-worked team members interact, collaborate and coordinate tasks and deadlines? What are the indicators of conflict and dissonance within a project team and how can they be identified, monitored and managed? How do project teams deal with and resolve dissonance or breakdowns in sensemaking within team and across distributed teams? and What obstructs and what stimulates collective sensemaking, collaboration and coordination?

The project management sensemaking level involves additional causes and sources of distress related to changing and often conflicting demands. Intra- and extra-project sensemaking overlap at this level. Conflicting demands often stem from the different stakeholders of a project (e.g. various types of users, project managers and project owners). Research questions of interest to ISD project management include the following: How do different stakeholders negotiate and resolve differences in ISD project objectives, funding, timeline and deadlines? How do (in)adequate funding and (un)realistic deadlines affect project performance and distress? How and why do perceptions and sensemaking of an ISD project within and external to a project team differ and what are the consequences for project managers? and How are these different perceptions recognized and addressed?

At the extra-project sensemaking level, numerous critical issues can lead to ISD project distress. At the governance structure level, the crucial question is how to align business strategy and IS strategy. Other relevant questions include the following: What are the implications of a rigid versus a flexible governance structure on ISD project performance and distress? How do IT governance structures 'control' and affect ISD projects and, conversely, how do ISD projects enact and challenge IT governance structures? What values, norms and rules underpin, on the one hand, IT governance structures and, on the other hand, ISD projects? What are the implications when these values, norms and rules are discordant with those embedded in ISD projects? How do differences between IT department culture and organizational culture emerge and affect ISD projects? What approaches, processes and interventions are used to productively deal with discordant values, norms and cultures?

The research questions for studying ISD project distress presented above are only illustrative. New questions will most likely emerge with systematic empirical investigations and deeper understanding of the ISD project distress phenomenon itself.

Discussion

In this article, we propose an extension from a focus on ISD project failure as a static end state to a focus on ISD

project distress as a dynamic and emergent state. Research focused on ISD project distress is proposed to examine troubling and harmful project conditions that cannot be routinely addressed, explained and resolved, and that increase the risk of project failure if left undetected and unattended to. The proposed extension of ISD project failure research by focusing on ISD project distress together with the suggested sensemaking framework for understanding and systematically exploring the nature and sources of distress, we hope, open up new avenues for research and creation of relevant knowledge. Focusing on a wide range of problematic situations and emergence of distress at and across different sensemaking levels during ISD project runtime would enable researchers to better understand sources and nature of distress and responses to such distress.

This extended focus on ISD project distress is warranted by a lack of deeper understanding of how complex problems emerge and create project distress and how distress lingers on and escalates or is identified and resolved in practice. As ISD project distress has not been systematically studied in the literature, we lack knowledge and skills to diagnose, deal with and alleviate it. While efforts have been made to formulate general (causal or process) models for ISD project failure (e.g. Biehl, 2007; Cooke-Davies, 2002; Lee and Xia, 2005; Lyytinen and Hirschheim, 1987; Wang et al., 2008), they have tended not to hold in most real-life situations due to the unique contexts of actual ISD projects (Davison and Martinsons, 2016). ISD project distress, as an alternative framing, lends itself more towards acknowledging the complex and contextual nature of signals, sources and dynamics of emerging problems and ways of problem identification, exploration and resolution of distress that are context-sensitive rather than fully generalized and abstracted. Process models in ISD project escalation literature, however, addressed some problems related to managers' decisions to continue or abandon a project (escalation or re-escalation) in real-life situations (Chulkov and Desai, 2008; Desai and Chulkov, 2009; Keil and Robey, 1999; Koster and Lambooj, 2018; Mähring and Keil, 2008; Montealegre and Keil, 2000; Pan and Pan, 2011). The concept of ISD project distress and the proposed sensemaking model of ISD project distress may contribute to process-oriented ISD project escalation literature by broadening their problem space so as to include project distress across the sensemaking levels that might reveal further uncertainty and complexity of managers' decisions.

ISD project distress as a framing corresponds well to the changing nature and increasing complexity of ISD projects. For example, outcome-oriented research on ISD project failure aligns with more traditional IS development approaches and methods such as linear 'waterfall' models with their rather clear deadlines and deliverables. The proposed ISD project distress research may be more

suitable for contemporary approaches such as agile development (for a review, Conboy, 2009). The framing of distress helps researchers to focus more on ongoing practices rather than final outcomes. Many contemporary IS development projects may not have a final outcome (may remain in ‘perpetual beta’) or definite project completion times (may be ‘continuously deployed’), hence making the focus on final outcomes both more difficult and less meaningful. Agile methods, it can be argued, focus more on short-term decision-making and skilful action, less on accurate upfront planning, which corresponds well to the concept proposed here.

In this article, we analyse, critically assess and reflect on the existing body of literature on ISD project failure and based on this analysis, develop this alternative framing of ISD project distress. The resulting analysis and contribution of the article, corresponding to our three objectives articulated in the introduction, can be summarized in three points.

First, we investigate the relevant literature and describe different research perspectives from which ISD project failure has been studied: the rationalist perspective, the process perspective, and the narrative and social constructivist perspectives. Their foci, assumptions, contributions and weaknesses are summarized in Table 1. While this rich body of literature provides a variety of understandings of ISD project failure, we have demonstrated that too much emphasis has been placed on failure as an end state and final assessment of an ISD project. What is missing is a focus on, and hence a more nuanced understanding of problematic situations that emerge during ISD projects.

Second, drawing from the critical assessment of and insights from the reviewed literature, we propose and develop the new concept of ISD project distress. We defined project distress as a troubling and harmful project condition that cannot be routinely resolved and that requires heedful attention and agile and flexible responses. ISD project distress is a dynamic and fluid constellation of critical problems that are typically not easily identified, understood and addressed. As a highly relevant and complex phenomenon, ISD project distress warrants focused and systematic investigations in many different real-life contexts. While we found studies that referred to problematic situations in ISD projects, the concept of ISD project distress has not been explicitly elaborated, mentioned or studied. By developing this concept, the article makes a contribution to knowledge and opens up a new domain of research that to date has been neglected in the literature.

Third, we have contributed a sensemaking theoretical framework for understanding and exploring ISD project distress (based on Weick, 1995) and proposed a related research agenda. The model distinguishes between three key sensemaking levels critical to ISD projects. At the intra-project sensemaking level, the relevant issues of concern relate to the individual team members’ sensemaking

and the occurrence of discord or breakdowns in their social interactions and collective sensemaking. Of future research interest at this level are the ways in which individual team members make sense of and negotiate project specifications, objectives and deliverables; the approach they take to collaborate and coordinate their tasks and deadlines; and the indicators of conflict and dissonance within a project team. At the project management level, where intra-project and extra-project sensemaking overlap, sources of distress are related to management approaches to ISD projects, including a variety of strategies to deal with the triple constraints (time, budget and scope) imposed on projects by an organization. Moreover, the discrepancy in ISD project perceptions among different stakeholders and, in particular, project owners and users is identified as an important cause of project distress that needs further investigation. The organizational, extra-project sensemaking processes are concerned with cultural, structural and governance issues involved in ISD project management. Some research questions at this level include the potential cultural (mis)fit between the information system being developed and the organizational values and traditions; the politics surrounding ISD project management, assessment and funds allocation; and connection between the prescribed development methodology and ISD practice. As the article demonstrates, the sensemaking theoretical framework allows the identification of a range of new research problems related to ISD project distress.

Future research, we argue, should contribute to a better understanding of organizational and project conditions and the nature, signals and significance of distress and should also seek effective interventions and organizational actions to deal with project distress. Such a research focus will enable new insights and valuable knowledge that might assist in turning around struggling projects and preventing failures. This article thus has important implications for both research and practice as it articulates novel research problems, a new concept of ISD project distress, a sensemaking theoretical framework and a research agenda that is relevant and valuable for real-life ISD projects.

Of course, the research questions articulated in the research agenda are only illustrative and are not intended to be exhaustive. However, they indicate the kind of research questions that arise when ISD research reorients to problematic situations and sources of project distress. The questions outline a research agenda grounded in the multilevel sensemaking theoretical framework as developed in this article. The full impact and nature of the research on ISD project distress cannot be fully assessed without such research.

Conclusion

In this article, we have asked these questions: What are the contributions and weaknesses of the extant ISD project failure/success literature? and What are potential avenues to

move the ISD literature forward? To answer, first, we provided a literature review and a critical assessment of the ISD project success/failure literature, analysing the assumptions, contributions and weaknesses of three different approaches that we identified. Second, we developed the concept of ISD project distress and provided an argument for an extension of research towards ISD project distress as a distinct focus. Third, we provided a sensemaking theoretical framework for in-depth and comprehensive exploration of ISD project distress and based on the framework, we suggested the research agenda. Overall, the article argues for extending the research focus from ISD project failure as an end state towards ISD project distress and the problematic situations experienced during ISD projects. This shift in focus in IS failure research towards early treatment of ISD project distress is expected to produce valuable knowledge of high relevance to practice.

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Notes

1. In this article, we use the expression 'ISD project failure' to refer to the literature on both ISD project failure and success, as studies often discuss both and conceptualize success as the opposite of failure.
2. Medical terminology has been found useful in the ISD literature and in practice. For example, terms such as 'symptoms' (Kappelman et al., 2006); 'health checks' (Ward, 2003); 'organizational pathology' (Goulielmos, 2003); and 'post-mortem reviews' (Kasi et al., 2008; Nelson, 2007) are common terminology in the study of ISD projects. The term 'distress' has also been used in other contexts such as 'a situation when a ship, aircraft etc. is in danger and needs help' (Longman Dictionary of Contemporary English, 2003).

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